

James Friedenber

Software Engineer | Frontend & Platform Systems

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[LinkedIn](#) | [Portfolio](#) | [Github](#)

SUMMARY

Software Engineer with experience building scalable live-event systems, developer tooling, client/server performance infrastructure, and modern web applications at Microsoft. Directly supported millions of players, large-scale live events, and creator ecosystems.

PROFESSIONAL EXPERIENCE

Software Engineer, Microsoft - Remote, USA

March 2022 - June 2026

- Co-engineered Minecraft: Bedrock Edition's first automated multiplayer performance testing framework, enabling player-per-server capacity increase from 10 to 80 players and reducing infrastructure for a 25k concurrent-player live event from ~2,500 to ~313 Azure VMs.
- Developed foundational gameplay and event systems, including Minecraft's first official in-game cutscene system, for the Minecraft Legends 2023 promotional live event, reaching over 3M players.
- Rebuilt Minecraft live events management tooling as a React web application, integrating it into the Auger website content pipeline to enable distribution to third party creators and streamline the live events deployment process.
- Built a telemetry framework for MCTools using Application Insights that supports custom event logging and real-time issue detection.
- Developed a localization framework, enabling MCTools to be available in 20+languages and bringing it into compliance with the minecraft.net domain.
- Modernized the MCTools TypeScript codebase and expanded content validation suites to reduce manual review workload for the Minecraft Vanilla team and external creators.

Software Design Engineer in Test, DIGARC - Remote, USA

March 2021 - March 2022

- Developed detailed test plans and tracked quality initiatives using Jira for an Agile team of developers, testers, and a PM.
- Created and maintained several hundred automated E2E tests, ensuring reliable frontend regression coverage.
- Led weekly team meetings to evaluate performance and functionality metrics in preparation for production deployments.

Software Design Engineer in Test, Andromeda Systems Inc. - Jacksonville, Florida

August 2019 - March 2021

- Developed over 500 automated tests using Selenium for an inventory management application used by the U.S. Navy for aircraft maintenance.
- Built dozens of Angular components using TypeScript for administrative tools to improve client operations and workflows.
- Ensured product quality in an Agile environment by triaging bugs and writing user stories and test cases.

QA Automation Engineer CO-OP, LenelS2 - Rochester, New York

January 2018 - August 2018

- Developed and designed a cloud-based facilities security product, contributing Selenium frontend automated tests and UI/UX improvements.
- Tested software interactions with security scanners by configuring and running automated tests in Linux terminals.
- Ensured product specifications met security standards and maintained quality by developing detailed test plans in an Agile environment.

SKILLS

Languages: TypeScript/JavaScript, C++, C#, Python

Web Development: React, Angular, Node.js, .NET, REST APIs, TailwindCSS, Zod, Zustand

Testing and Automation: Playwright, Selenium, React Testing Library, Jest, Postman, Swagger

Cloud and Infra: Git, Azure, Jenkins, SQL, PostgreSQL, CI/CD

AI-Assisted Development: GitHub Copilot, Large Language Models (LLMs), Generative AI, OpenAI API, Retrieval-Augmented Generation (RAG), Vector Databases

EDUCATION

Rochester Institute of Technology - BS in Game Design and Development

August 2015 - May 2019